

# FILTER DESIGN TEST CASE

Band-Pass Filter - Band 28 (700 MHz) for WiFi 6E Coexistence Filter  
Design Validation and Performance Characterization

## DOCUMENT INFORMATION

|                  |                              |
|------------------|------------------------------|
| Document Number: | FD-TC-0095                   |
| Revision:        | Rev C.6                      |
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| Reviewer:        | Ana Reyes                    |
| Design Center:   | San Jose Advanced Design Lab |
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| Status:          | PASS                         |
| Priority:        | P1 - Critical                |

## 1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting Band 28 (700 MHz) for WiFi 6E Coexistence Filter. The filter is designed on PZT on Si substrate with a target center frequency of 1758.1 MHz and bandwidth of 357.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

## 2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: Band 28 (700 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: PZT on Si
- Package: Fan-Out WLP
- Design Tool: Keysight ADS 2024
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Band-Pass Filter design targeting Band 28 (700 MHz). Results are used to qualify the design for mass production release.

## 3. DESIGN PARAMETERS

|                          |                |
|--------------------------|----------------|
| Center Frequency:        | 1758.1 MHz     |
| Bandwidth (3 dB):        | 357.6 MHz      |
| Insertion Loss Target:   | $\leq 2.3$ dB  |
| Return Loss Target:      | $\geq 16.0$ dB |
| Out-of-Band Rejection:   | $\geq 36.0$ dB |
| Isolation (Duplexer):    | $\geq 20.0$ dB |
| Q Factor:                | 1510           |
| Temperature Coefficient: | -17.5 ppm/°C   |
| Die Size:                | 1.5 x 0.7 mm   |
| Wafer Size:              | 6-inch         |
| Number of Resonators:    | 4              |
| Electrode Material:      | W              |

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the Band-Pass Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 1758.1 MHz, BW 357.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 5274 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.3 dB
- Return loss in passband shall be >= 16.0 dB
- Out-of-band rejection at +/- 536 MHz offset >= 36.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 3.8 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 86%

6. TEST RESULTS

Measurement Date: 2024-10-11  
Devices Tested: 17  
Insertion Loss (meas): 2.01 dB  
Return Loss (meas): 17.7 dB  
Rejection (meas): 38.5 dB  
Isolation (meas): 25.1 dB  
Q Factor (meas): 1510  
Group Delay Variation: 5.8 ns  
Power Handling: +35 dBm  
Die Yield: 98.7%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Band-Pass Filter design demonstrated excellent performance for Band 28 (700 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

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No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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|-----------|---------------|-------|------------|
| Author:   | Jennifer Kim  | Date: | 2024-10-11 |
| Reviewer: | Ana Reyes     | Date: | 2024-10-16 |
| Approver: | Dr. James Liu | Date: | 2024-10-18 |

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